
Table of Contents

MATLAB Project: a least squares project for house price models	1
Example --- making a scatter plot	1
MATLAB question 1 --- loading and visualizing housing data	2
Pen and paper question 1	4
MATLAB question 2 --- least squares modeling of housing data	4
Pen and paper question 2	5

MATLAB Project: a least squares project for house price models

In this assignment, you will explore data using scatter plots and solve a least squares problem involving real-world data. The real data example is taken from Boyd and Vandenberghe's "Introduction to Linear Algebra".

Place the requested computations for the MATLAB questions in this file and use MATLAB's publish feature to produce a PDF of the results.

- go to the PUBLISH tab and click the arrow under the Publish button and select Edit Publishing Options
- In "Output Setting" within publishing options, choose Output File Format to be pdf.
- Close the options window and press the play button labeled "Publish"

Be sure to use semi-colons to prevent unwanted displays of large matrices/vectors in the PDF. You should also verify that the requested outputs (mostly plots but some numbers as well) are visible in the produced PDF.

There are 2 MATLAB questions and 2 pen and paper questions to answer. No work is required from you in the "Example" sections. These are provided to give you a sense of how to use the appropriate functions. Complete the pen and paper questions and submit them as a separate PDF.

```
clearvars; clc;
```

Example --- making a scatter plot

A scatter plot is useful for visualizing data where it doesn't necessarily make sense to connect the dots, as in a plot. MATLAB's scatter function allows you to make scatter plots. Below we show how to make a scatter plot for some randomly generated data, where we set marker size based on data. Given x-coordinates in a vector X, y-coordinates in a vector Y, and marker sizes in a vector S, we can make the scatter plot using

```
scatter(X,Y,S)
```

The units for marker size are in points².

```
% in the made up example below we have some (X,Y) data given by
random,
% noisy sampling of a cosine function. We set marker sizes based on
the
% amplitude (abs(Y)) and color based on the phase (X restricted
between
```

```

% 0 and 2*pi).

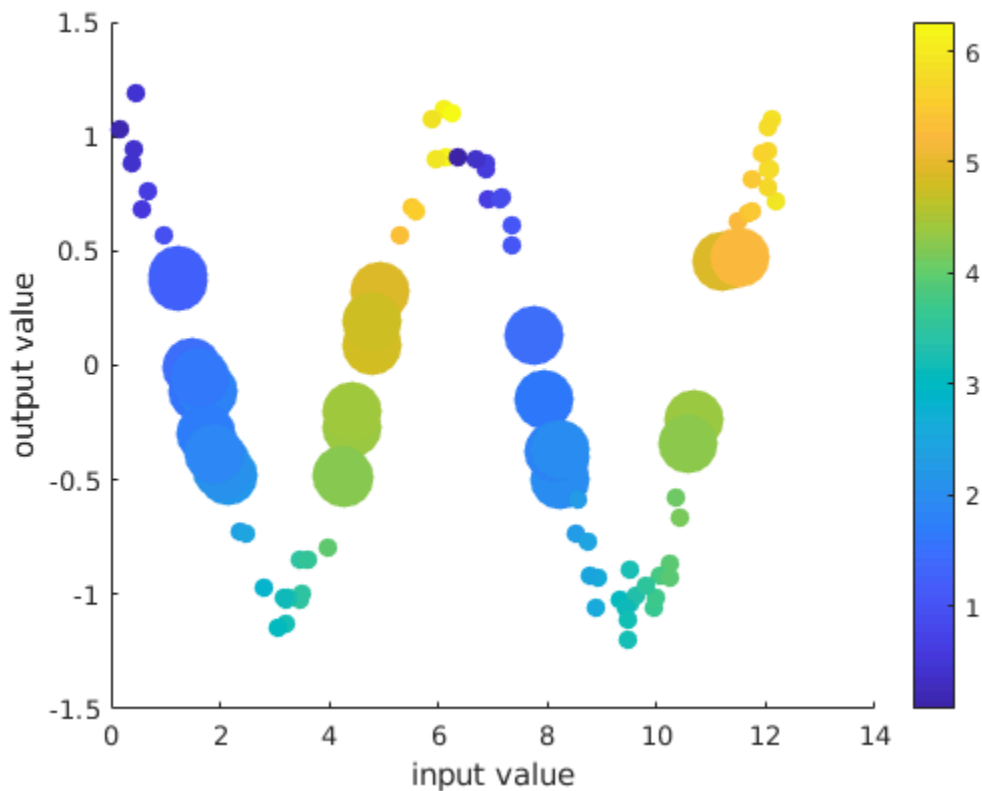
figure(1)
clf
n = 100; % number of samples
sigma = 0.1; % noise level
X = 4*pi*rand(n,1); % random samples in [0,4*pi]
Y = cos(X)+randn(n,1)*sigma; % noisy output values for cos(X)
S = 50*(abs(Y) > 0.5) + 500*(abs(Y) <= 0.5); % change marker size
based

                                % on amplitude
C = X-floor(X/(2*pi))*2*pi; % set color based on phase of X
scatter(X,Y,S,C,'filled') % 'filled' gives filled in markers,
otherwise

                                % only the outline is drawn
xlabel('input value') %sets a label for x-axis
ylabel('output value') %sets a label for y-axis

colorbar % shows what values the colors correspond to

```



MATLAB question 1 --- loading and visualizing housing data

The function `house_sales_data` returns a matrix whose columns contain house sales information for 774 homes in the Sacramento area. You can call this function using:

```
H = house_sales_data();
```

The columns of the output matrix H are as follows.

- Column 1: a region number (1--4) based on 5-digit ZIP code
- Column 2: number of bedrooms.
- Column 3: number of bathrooms.
- Column 4: house area in square feet.
- Column 5: 1 if a condo, 0 otherwise.
- Column 6: selling price in dollars.

We would like to explore this data a little using scatter plots.

One variable we are interested in predicting is the selling price. A major driver of selling price is house area.

----- Instructions for question 1 -----

Below, first load the housing data into a matrix and then make a scatter plot where

- X = house area
- Y = selling price
- S = 5*(number of bedrooms)^2
- C = region number

The X,Y,S,C are x-coord, y-coord, marker size, and marker color as in the example above.

Label the X and Y axes accordingly

Also, make 2 additional plots where

- X = number of bedrooms
- Y = selling price
- C = number of bathrooms

and

- X = house area
- Y = number of bedrooms
- C = number of bathrooms

if S or C is not specified it can be ignored, in that case you can use

```
scatter(X,Y, 'filled')
scatter(X,Y,S, 'filled')
scatter(X,Y, [],C, 'filled')
```

As appropriate.

Again, label the X and Y axes accordingly

% put your code for MATLAB Question 2 here

Pen and paper question 1

Comment (in about 5 sentences) on the patterns and trends you observe in the scatter plots

MATLAB question 2 --- least squares modeling of housing data

Suppose we would like to predict a home sale price in Sacramento using information like the home's region, number of bedrooms, number of bathrooms, house area, and whether or not it's a condo.

A very basic model would take the home price prediction to be a linear combination of these variables. We can use the housing data to obtain reasonable values for the coefficients in this linear model as the solution of a least squares problem.

Define a matrix A to be made up of columns 2,3,4, and 5 (no. of beds, no. of baths, house area, and condo or not) of the housing data and a vector b given as the 6th column (house price). Then, find the coefficients c as the solution of the least squares problem:

$$c = \operatorname{argmin}_c \|Ac - b\|$$

Recall that the MATLAB backslash operation will compute this solution when A is rectangular, i.e. you can obtain c by

$$c = A \backslash b;$$

These coefficients can also be computed given a QR decomposition. If $A=QR$ then

$$c = R^{-1}Q^T b$$

A fast and stable method for QR is available as a built-in in MATLAB. It can be called as follows:

$$\begin{aligned} [Q, R] &= \operatorname{qr}(A); \\ [Q, R] &= \operatorname{qr}(A, \theta); \end{aligned}$$

The second call produces a "reduced" QR, which is sufficient in most cases including solving least squares problems.

----- Instructions for question 2 -----

Below, form A and b as described and compute vectors of coefficients c_1 and c_2 using the backslash operation and a QR decomposition, respectively. Compare c_1 and c_2 numerically and print their values out.

One way to think about the quality of a linear model is to compute the relative residual

$$\frac{\|Ac - b\|}{\|b\|}$$

Compute this value for both $c=c_1$ and $c=c_2$ and display the values below.

Finally, try fitting a simpler linear model using only number of bedrooms and house area for the columns of A. Print out the coefficients and relative residual for this model as well.

`% put your matlab code for MATLAB question 3 here.`

Pen and paper question 2

- Explain why we would not include something like the region ID in a linear model.
- How do the relative residuals between the larger and smaller linear models compare?
- Comment on whether or not the coefficients obtained for the smaller model (just house area and number of bedrooms) make sense in context (for instance, are the coefficients positive which you think should be positive?).
- Observe that house area and number of bedrooms are correlated. Does that help interpret the results?

Published with MATLAB® R2021a